

* We recommend adapting this protocol to use the Gene Pulser electroporation buffer (catalog #165-2676, 165-2677), which increases cell viability and transfection efficiency in mammalian cell lines.

Cell Type	Mammalian, suspension	Molecules Electroporated	DNA: variety of constructs, supercoiled, usually <10 kB (some pUC based)
Species Used	Mouse, erythroleukemia cells		

Before the Pulse

Cell Growth Medium	DMEM + 10% Fetal Calf Serum, 1 x glutamine, 1 x penicillin / streptomycin (GIBCO/BRL, Sigma)	Growth Phase at Harvest	Log phase
Wash Solution	DMEM + 10% Fetal Calf Serum, 1 x glutamine	Pre-pulse Incubation	10 min., room temperature

The Pulse

Instruments Used Gene Pulser® apparatus & Capacitance

Electroporation Temperature	Room temperature	Cuvette Gap	0.4 cm
Electroporation Medium*	DMEM + 1% Fetal Calf Serum	Voltage	0.250 kV
Cell Density	1.43 x 10 ⁷ cells / ml	Field Strength	0.625 kV/cm
Volume of Cells	700 µl	Capacitor	960 µF
DNA Concentration	20 to 30 µg /100 µl	Resistor	(Pulse Controller) Ω none
DNA Resuspension Buffer	10 mM Tris, pH 8.0, 1 mM EDTA	Time Constant	16 msec (some small variation)
Volume of DNA	100 µl		
After the Pulse			
Outgrowth Medium	DMEM + 10% Fetal Calf Serum, 1 x glutamine, 1 x penicillin / streptomycin		

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.

Outgrowth Temperature	37 °C
Length of Incubation	24 to 48 hours
Selection Method or Assay Used	CAT assay
Electroporation Efficiency	Not given
Per Cent Survival	35%

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